

111.
1.

Given Relations, EMP (ENO, ENAME, TITLE)
ASG (ENO, PNO, RESP, DUR)

Query: Find the names of employees who are managing a project.

Step 1: Select managers from ASG

$\sigma_{RESP='Manager'}(ASG)$

Step 2: Join with EMP

$EMP \bowtie_{EMP.ENO=ASG.ENO} \sigma_{RESP='Manager'}(ASG)$

Step 3: Project employee names

$\pi_{ENAME}(EMP \bowtie_{EMP.ENO=ASG.ENO} \sigma_{RESP='Manager'}(ASG))$

2.

Given Schemas, EMP (ENO, ENAME, TITLE)
PROJ (PNO, PNAME, BUDGET)
ASG (ENO, PNO, RESP, DUR)

Step 1: Select the CAD/CAM project

$\sigma_{PNAME='CAD/CAM'}(PROJ)$

Step 2: Select employees who worked for 1 or 2 years

$\sigma_{DUR=1 \text{ OR } DUR=2}(ASG)$

Step 3: Join PROJ and ASG

$\sigma_{PNAME='CAD/CAM'}(PROJ) \bowtie_{PROJ.PNO=ASG.PNO} \sigma_{DUR=1 \text{ OR } DUR=2}(ASG)$

Step 4: Join ^{with} EMP

$EMP \bowtie_{EMP.ENO=ASG.ENO} [\sigma_{PNAME='CAD/CAM'}(PROJ) \bowtie_{PROJ.PNO=ASG.PNO} \sigma_{DUR=1 \text{ OR } DUR=2}(ASG)]$

$\sigma_{ENAME \neq 'J.Doe'}(\dots)$

Step 5: Exclude J.Doe.

$\sigma_{ENAME \neq 'J.Doe'}(\dots)$

$\pi_{ENAME} (\sigma_{ENAME \neq 'J.Doe'} (EMP) \bowtie_{ENO} (\sigma_{PNO} [\sigma_{PNAME = 'CAD/CAM'} (PROJ) \bowtie_{DUR=1 \text{ OR } DUR=2} (ASG)]))$.

3. Query decomposition is the process of breaking a complex query into smaller and simpler operations so that the query can be processed efficiently.

In a distributed database, query decomposition is an important step because the data may be stored at different sites.

Example: Consider $EMP(ENO, ENAME, TITLE)$

$ASG(ENO, PNO, RESP, DUR)$

Query: Find the names of employees who are working as managers.

SQL Query:- Select ENAME From EMP, ASG where EMP.ENO = ASG.ENO

AND ASG.RESP = 'Manager';

Query Decomposition:

Step 1: select managers: $\sigma_{RESP = 'Manager'} (ASG)$

Step 2: Join with Emp: $EMP \bowtie_{EMP.ENO = ASG.ENO} \sigma_{RESP = 'Manager'} (ASG)$

Step 3: Select employee names:

$\pi_{NAME} (EMP \bowtie_{EMP.ENO = ASG.ENO} \sigma_{RESP = 'Manager'} (ASG))$

This is optimized query in relational algebra.

4. What is semi join?

A semi join is a distributed join technique used to reduce the amount of data transferred b/w sites. $R \ltimes S = R \ltimes_{\pi_A} (S)$

Steps of semi join:

① Project the join attribute: $\pi_A(R) = \{1, 2, 4, 5\}$

② Send these values from site 1 $\rightarrow$ site 2.

③ select matching tuples from S: $S' = \{(1, 4), (2, 5)\}$

④ Send reduced S' to site 1.

⑤ Perform the final join: $R \bowtie S'$

A	C	B
1	6	4
2	7	5

Cost Analysis:

- Ordinary Join: Transfer complete $R = 4$
- Semi Join: Transfer R 's A tuples = 4 tuples + reduced $S = 2$ tuples

Total semijoin Transfer = 6 tuples.

5.

Given: R is the outer relation at site 1.

S is the inner relation at site 2.

Join condition: $R.A = S.A$

Four Strategies of R^* Algorithm:

a) Move Outer Relation to Inner Site: Move R from site 1 to site 2 and perform the join there. $R \rightarrow \text{site 2}$

Suitable where R is small.

b) Move Inner Relation to Outer Site: Move S from site 2 to site 1 and perform the join there. $S \rightarrow \text{site 1}$

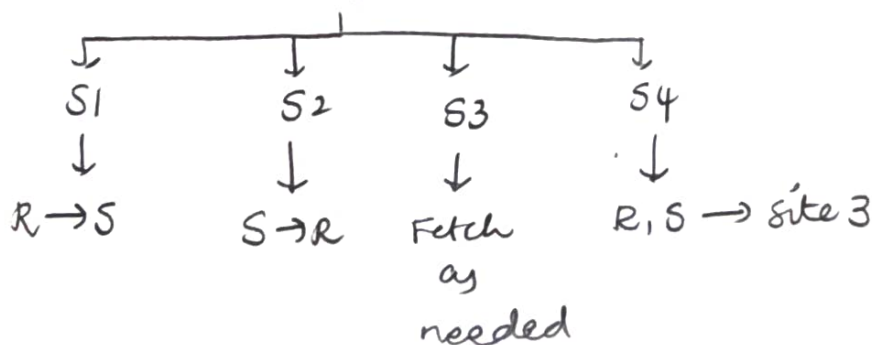
Suitable where S is small.

c) Fetch Inner Tuples as Needed: For each key tuple of R , send its join attribute to site 2. Site 2 finds the matching tuples of S and sends only those tuples back. $R \rightarrow A \text{ values} \rightarrow S$

d) Move Both Relations to a Third site: Move both R and S to another site and perform the join there. $R, S \rightarrow \text{site 3}$

Suitable when a third site provides a lower cost overall.

R^* Algorithm





DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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UNIT II ASSIGNMENT-2 QUESTIONS

I. Multiple Choice Questions (MCQs)

1. What is query processing?
A) Storing data B) Executing a query efficiently
C) Creating tables D) Deleting data [B]
2. The main objective of query processing is to:
A) Increase data redundancy B) Minimize query execution cost
C) Increase storage space D) Delete duplicate data [B]
3. Which of the following is a layer of query processing?
A) Query decomposition B) Data deletion
C) File formatting D) User authentication [A]
4. Which operation identifies the fragments required for a query?
A) Localization B) Encryption C) Compression D) Sorting [A]
5. Distributed query optimization considers:
A) Data location B) Communication cost
C) Processing cost D) All of the above [D]
6. Which technique can reduce the amount of data transferred between sites?
A) Semijoin B) Cartesian product C) Duplication D) Encryption [A]
7. A query processor is responsible for:
A) Processing database queries B) Designing hardware
C) Creating operating systems D) Managing printers [A]
8. A query execution plan specifies
A) How a query will be executed B) User passwords
C) Database users only D) Hardware configuration [A]

II. FILL IN THE BLANKS

1. Query decomposition is commonly expressed using relation algebra.
2. The process of identifying the required data fragments is called data localization.
3. In a distributed database, data can be stored at multiple sites.
4. The process of selecting the best execution plan is called query optimization.
5. Distributed query optimization considers communication cost.
6. The best execution plan generally has the less total cost.
7. Query optimization improves query performance.
8. Centralized query optimization is performed at a central site.

III Write all Questions Answer

1. Transformation of an SQL-query into an Relational Alegebra(RA)-query.

Relations: EMP(ENO, ENAME, TITLE), ASG(ENO,PNO,RESP,DUR)

Query: *Find the names of employees who are managing a project?*

2. Convert relational calculus query to relational algebra query and find an efficient expression.

Query: Find the names of employees other than J. Doe who worked on the CAD/CAM project for either 1 or 2 years.

Schemas: EMP(ENO, ENAME, TITLE)

PROJ(PNO, PNAME, BUDGET)

ASG(ENO, PNO, RESP, DUR)

3. Explain the following layers with diagram.

- a) Query Decomposition
- b) Data Localization
- c) Global Optimization
- d) Local Optimization

OR

Explain Query Decomposition with example.